WORKED SOLUTION

STEP II 2012, Question 3

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q3

The key idea

The awkward expression $x+\sqrt{1+x^2}$ is crying out to be called t. The trick that makes the substitution work is that its reciprocal is $\sqrt{1+x^2}-x$, so adding and subtracting t and $1/t$ gives both x and $\sqrt{1+x^2}$ as simple functions of t. After that, each of the two integrals is a matter of spotting how to write the integrand as a function of $x+\sqrt{1+x^2}$ alone.

Solution

The identity

Let $t=x+\sqrt{1+x^2}$. Multiplying by the conjugate,

$$\frac{1}{t}=\frac{1}{\sqrt{1+x^2}+x}\cdot\frac{\sqrt{1+x^2}-x}{\sqrt{1+x^2}-x}=\frac{\sqrt{1+x^2}-x}{(1+x^2)-x^2}=\sqrt{1+x^2}-x.$$

Subtracting and adding these two equations gives

$$x=\tfrac{1}{2}\Big(t-\frac{1}{t}\Big),\qquad \sqrt{1+x^2}=\tfrac{1}{2}\Big(t+\frac{1}{t}\Big).$$

The first of these gives x directly as a function of t, so

$$\mathrm{d}x=\tfrac{1}{2}\Big(1+\frac{1}{t^2}\Big)\mathrm{d}t.$$

For the limits: when $x=0$, $t=1$; and as $x\to\infty$, $t\to\infty$. Also t is a strictly increasing function of x (both x and $\sqrt{1+x^2}$ increase for $x\geq 0$), so the substitution is one-to-one on $[0,\infty)$. Therefore

$$\int_0^\infty f(x+\sqrt{1+x^2})\,\mathrm{d}x=\frac{1}{2}\int_1^\infty\Big(1+\frac{1}{t^2}\Big)f(t)\,\mathrm{d}t,$$

as required.

The first integral

Ask yourself How can the denominator $2x^2+1+2x\sqrt{x^2+1}$ be a function of $x+\sqrt{1+x^2}$?

Try squaring: $(x+\sqrt{1+x^2})^2 = x^2+2x\sqrt{1+x^2}+(1+x^2) = 2x^2+1+2x\sqrt{x^2+1}$, which is exactly the denominator. So the integrand is $f(x+\sqrt{1+x^2})$ with $f(t)=1/t^2$, and the identity gives

$$\int_0^\infty \frac{\mathrm{d}x}{2x^2+1+2x\sqrt{x^2+1}} = \frac{1}{2}\int_1^\infty \Big(1+\frac{1}{t^2}\Big)\frac{1}{t^2}\,\mathrm{d}t = \frac{1}{2}\int_1^\infty (t^{-2}+t^{-4})\,\mathrm{d}t$$
$$= \frac{1}{2}\Big[-\frac{1}{t}-\frac{1}{3t^3}\Big]_1^\infty = \frac{1}{2}\Big(1+\frac{1}{3}\Big) = \frac{2}{3}.$$

The second integral

Put $x=\tan\theta$, so $\mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta$, that is, $\mathrm{d}\theta = \dfrac{\mathrm{d}x}{1+x^2}$. As θ runs from 0 to $\frac{1}{2}\pi$, x runs from 0 to ∞. Since $0\le\theta<\frac{1}{2}\pi$, $\cos\theta>0$, so $\sec\theta=\sqrt{1+x^2}$ and

$$\sin\theta = \tan\theta\cos\theta = \frac{x}{\sqrt{1+x^2}}, \qquad 1+\sin\theta = \frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}} = \frac{t}{\sqrt{1+x^2}},$$

where again $t=x+\sqrt{1+x^2}$. Hence

$$\int_0^{\frac{1}{2}\pi}\frac{\mathrm{d}\theta}{(1+\sin\theta)^3} = \int_0^\infty \frac{(1+x^2)^{3/2}}{t^3}\cdot\frac{\mathrm{d}x}{1+x^2} = \int_0^\infty \frac{\sqrt{1+x^2}}{t^3}\,\mathrm{d}x.$$

Ask yourself There's still a $\sqrt{1+x^2}$ in the integrand. How do we write it in terms of t?

We found it in the first part: $\sqrt{1+x^2}=\frac{1}{2}(t+1/t)$. So the integrand is

$$\frac{t+t^{-1}}{2t^3} = \tfrac{1}{2}(t^{-2}+t^{-4}),$$

a function of $t=x+\sqrt{1+x^2}$ only. Applying the identity with $f(t)=\frac{1}{2}(t^{-2}+t^{-4})$,

$$\int_0^{\frac{1}{2}\pi}\frac{\mathrm{d}\theta}{(1+\sin\theta)^3} = \frac{1}{2}\int_1^\infty\Big(1+\frac{1}{t^2}\Big)\cdot\frac{1}{2}\Big(\frac{1}{t^2}+\frac{1}{t^4}\Big)\mathrm{d}t = \frac{1}{4}\int_1^\infty (t^{-2}+2t^{-4}+t^{-6})\,\mathrm{d}t$$
$$= \frac{1}{4}\Big(1+\frac{2}{3}+\frac{1}{5}\Big) = \frac{1}{4}\cdot\frac{28}{15} = \frac{7}{15}.$$

Checks

- Test the identity on a simple f: with $f(t)=1/t^3$ the right-hand side is $\frac{1}{2}\int_1^\infty(t^{-3}+t^{-5})\,\mathrm{d}t = \frac{1}{2}(\frac{1}{2}+\frac{1}{4}) = \frac{3}{8}$, and numerical quadrature of the left-hand side also gives 0.375.

- Simpson's rule on the original integrals gives 0.666667 and 0.466667, matching $\frac{2}{3}$ and $\frac{7}{15}$.
- Sanity check on size: on $[0, \frac{1}{2}\pi]$ the integrand $1/(1+\sin\theta)^3$ falls from 1 to $\frac{1}{8}$, so the integral lies between $\frac{1}{8}\cdot\frac{1}{2}\pi \approx 0.20$ and $\frac{1}{2}\pi \approx 1.57$. The value $\frac{7}{15} \approx 0.47$ is comfortably inside.

Common slips

- Trying to find $\mathrm{d}t/\mathrm{d}x$ and then express it in terms of t the hard way. Writing x as a function of t first and differentiating that is much quicker.
- In the second integral, forgetting the factor $1/(1+x^2)$ from $\mathrm{d}\theta$, or leaving $\sqrt{1+x^2}$ in the integrand rather than converting it to $\frac{1}{2}(t+1/t)$.

Beyond STEP

The substitution $t = x + \sqrt{1+x^2}$ is really $t = \mathrm{e}^u$ where $x = \sinh u$: indeed $\sinh^{-1} x = \ln\left(x+\sqrt{1+x^2}\right)$, and the formulas $x = \frac{1}{2}(t-1/t)$, $\sqrt{1+x^2} = \frac{1}{2}(t+1/t)$ are just $\sinh u$ and $\cosh u$ written with exponentials. Hyperbolic substitutions like this are a standard first-year tool for integrals involving $\sqrt{1+x^2}$. Siklos, *Advanced Problems in Mathematics*, has more STEP integrals in this spirit.

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